Tangente Lösungen · mehr

1.	**$g(t) \approx -0{,}597t + 93{,}77$; ab 90 Tagen**	$m = h'(60) = -12e^{-3} \approx -0{,}597$; $h(60) \approx 57{,}92 \Rightarrow g(t) \approx -0{,}597t + 93{,}77$; $g(t) \leq 40 \Rightarrow t \geq 90 \Rightarrow$ ab 90 Tagen
2.	**$x \approx 4{,}44$ ($(2 - 8\sqrt{2})/3 < 2$ entfällt)**	$f'(x) = -\frac{3}{50}x^2 + \frac{2}{25}x + \frac{2}{5} = -\frac{32}{75} \Rightarrow 9x^2 - 12x - 124 = 0 \Rightarrow x = (2 \pm 8\sqrt{2})/3 \Rightarrow x = (2 + 8\sqrt{2})/3 \approx 4{,}44$ ($(2 - 8\sqrt{2})/3 < 2$ entfällt)
3.	**t: $y = -2x + 4$**	$f(0) = 4$; $f'(0) = -2 \Rightarrow$ t: $y = -2x + 4$
4.	**t: $y = 2x + 1$**	$r(0) = 1 = s(0) \Rightarrow$ Punkt $(0\|1)$; $r'(x) = -2x + 2 \Rightarrow r'(0) = 2$; $s'(x) = 2e^{2x} \Rightarrow s'(0) = 2 \Rightarrow$ gleiche Steigung $\Rightarrow$ t: $y = 2x + 1$